

Q1. State Space Representation (Case Study) A logistics company is trying to optimize its delivery system in a smart city. The city is represented as a graph where nodes represent locations and edges represent routes with associated costs (distance, traffic, fuel consumption). The system must determine the most efficient route for delivery vehicles while considering constraints like traffic congestion and delivery deadlines. Tasks:

- 1.) Model the problem using state space representation (define state, operators, initial state, goal state).
- 2.) Explain how search algorithms like Breadth-First Search, Depth-First Search, and A* can be applied.
- 3.) Compare uninformed vs informed search strategies in this

context.

4.) Suggest the most suitable approach and justify your answer.

Ans.) 1. State Space Model

State: A state represents the current location of the delivery vehicle + remaining deliveries + current time/traffic condition. Example: (current_location, pending_deliveries, time)

Operators (Actions): Move from one node (location) to another via available routes (edges).

Initial State: Vehicle at depot with all deliveries pending.

Goal State: All deliveries completed within deadlines (optimal route

found).

Cost Function: Combination of distance, fuel consumption, and traffic delay.

Applying Search Algorithms

Breadth-First Search (BFS): Explores level by level.

Guarantees shortest path (in terms of steps, not cost).

Not suitable here because costs vary (traffic, fuel).

Depth-First Search (DFS): Explores deep paths first.

Low memory

usage

May find suboptimal or very poor routes.

A* (Best choice among these):

Uses: actual cost so far : heuristic
(e.g., estimated distance to destination)

Efficiently finds optimal routes using heuristics like:

straight-line distance

predicted traffic

3. Uninformed vs Informed Search

Aspect

Uninformed (BFS, DFS)

Informed (A*)

Knowledge

No domain

Knowledge
Uses heuristics

Efficiency
Low
High
Optimality

Limited
High (if heuristic admissible)

Use Case
Simple graphs
Real-world routing

4. Best Approach

A* is most suitable Why?
Handles weighted costs (distance + traffic)
Scalable for smart cities
Can integrate real-time data (traffic)

Q2. Reinforcement Learning (Case Study) A self-driving car company is developing an AI agent to learn how to drive safely in urban environments. The car receives rewards for safe driving and penalties for collisions, traffic violations, or inefficient routes. Tasks:

- 1.) Formulate the problem as a Reinforcement Learning (RL) model (states, actions, rewards, environment).
- 2.) Explain how Q-learning or Deep Reinforcement Learning can be used.
- 3.) Discuss the exploration vs exploitation trade-off in this scenario.
- 4.) Identify challenges such as sparse

rewards, safety concerns, and real-world deployment.

Ans.)

1. RL Model

States: Car position, speed, nearby vehicles, traffic signals

Actions: Accelerate, brake, turn, change lane

Rewards:

Safe driving

Reaching destination quickly —

Collisions — Traffic violations — Fuel inefficiency

Environment: Urban road system with dynamic traffic

2. Q-Learning / Deep RL

Q-Learning: Updates value:

Deep RL (DQN): Uses neural networks to

approximate Q-values Useful for:
large state spaces vision-based driving
(camera input)

3. Exploration vs Exploitation

Exploration: Try new actions (riskier but learning)

Exploitation: Use known best actions (safer)

In self-driving:

Too much exploration = dangerous

Solution: safe exploration (simulation training first)

4. Challenges

Sparse rewards (few success signals)

Safety-critical decisions

Real-world unpredictability

Sim-to-real transfer

problem

Q3. Neural Network Learning & Genetic Algorithms (Case Study) A healthcare organization wants to develop a predictive system for early detection of heart disease. They are considering two approaches: Neural Networks and Genetic Algorithms.

Tasks:

- 1.) Explain how a Neural Network model can be designed for prediction (architecture, training, backpropagation).
- 2.) Describe how Genetic Algorithms (GA) can optimize feature selection or model parameters.
- 3.) Compare both approaches in terms

of accuracy, interpretability, and computational cost.

4.) Suggest a hybrid approach combining Neural Networks and Genetic Algorithms.

Ans.) 1. Neural Network Design
Input Layer: Patient data (age, BP, cholesterol)
Hidden Layers: Feature extraction
Output Layer: Probability of heart disease

Training:
Loss function (e.g., cross-entropy)
Backpropagation updates weights

2. Genetic Algorithms (GA)
Used for: Feature selection (best medical indicators)
Hyperparameter

tuning

Steps:

Initialize population (feature subsets)

Evaluate fitness (model accuracy)

Selection

Crossover

Mutation

Repeat

3. Comparison

Aspect

Neural Networks

Genetic Algorithms

Accuracy

High

Moderate

Interpretability

Low

Medium

Speed

Fast after

training
Slower

Use
Prediction
Optimization

4. Hybrid Approach

Combine both:
GA selects best features
Neural Network performs prediction

Benefits:
Improved accuracy
Reduced overfitting
Efficient model

Q4. Natural Language Processing (NLP)
(Case Study) A multinational company wants to build a chatbot that can handle customer queries in multiple languages and understand user

sentiment.

Tasks:

- 1.) Explain how NLP techniques like tokenization, stemming, and parsing can be used.
- 2.) Describe the role of language models and transformers in chatbot development.
- 3.) Discuss challenges like ambiguity, sarcasm, and multilingual processing.
- 4.) Propose a system architecture for the chatbot.

Ans.) 1. NLP Techniques

Tokenization: Split text into words

Stemming: Reduce words (e.g., "running" → "run")

Parsing: Understand grammar

structure

2. Language Models & Transformers

Modern chatbots use transformers

Examples:

BERT (understanding)

GPT (generation)

Advantages:

Context awareness

Multilingual capability

Better conversation flow

3. Challenges

Ambiguity ("bank" = river or finance)

Sarcasm detection

Multiple languages

Cultural context differences

4. Proposed Architecture

Input processing (speech/text)

Language detection

NLP preprocessing

Transformer model

Sentiment analysis

module

Response generation

Output (text/voice)

Q5. Machine Learning (ML) (Case Study)

An e-commerce company wants to improve its recommendation system to increase sales and customer

satisfaction. Tasks: Identify the type of ML approach (supervised, unsupervised, reinforcement learning). Explain how collaborative filtering and content-based filtering can be applied.

Discuss data preprocessing, feature engineering, and evaluation metrics.

Ans.) ML Approach

Supervised Learning: Predict ratings

Unsupervised Learning: Customer segmentation

Reinforcement Learning: Adaptive recommendations

Most systems use hybrid